

Sexual Anatomy and Physiology

Ms. Mitchell



Sexual Anatomy

Anatomy — the study of the physical structure and systems

Physiology — the study of how body systems and structures function

Comfort with your body supports communication and healthy decision-making





Sexual Anatomy

Genitals, or genitalia — the sexual and reproductive organs of both men and women.

Function of female and male sex organs:

- Reproduction
 - Today's culture has shifted this view toward sexual pleasure
- 



The Endocrine System and Hormones

Endocrine system — glands that regulate body functions by releasing hormones

- Controls puberty, metabolism, and mood

Hormones — chemical produced in the body that regulates cell or organ activity





The Endocrine System and Hormones

Steroid hormones — androgens and estrogens

- **Testosterone** — male reproduction, secondary sex traits, sexual function
- **Estrogens** — vaginal lubrication, wall maintenance, well-being
- **Progesterone** — uterine preparation and maintenance

Gonads— glands that produce sex hormones and reproductive cells

- Testes in the male, ovaries in the female.
- 



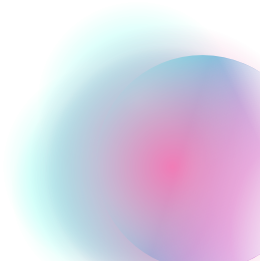
The Endocrine System and Hormones

Gonadotropin-releasing hormone (GnRH) — pituitary hormone that stimulates gonad activity; acts directly on the gonads

Luteinizing hormone (LH) — secreted by the pituitary

- Females: stimulates ovulation
- Males: stimulates androgen production in the testes

Follicle-stimulating hormone (FSH) — secreted by the pituitary

- Females: regulates ovulation and stimulates follicle development
 - Males: stimulates sperm production
- 



The Endocrine System and Hormones

Neuropeptide hormones — influence emotional aspects of sexuality, including attraction and arousal

Oxytocin — key neuropeptide affecting erotic attraction and love

- Facilitates milk flow during breastfeeding
 - Helps form emotional bonds
- 

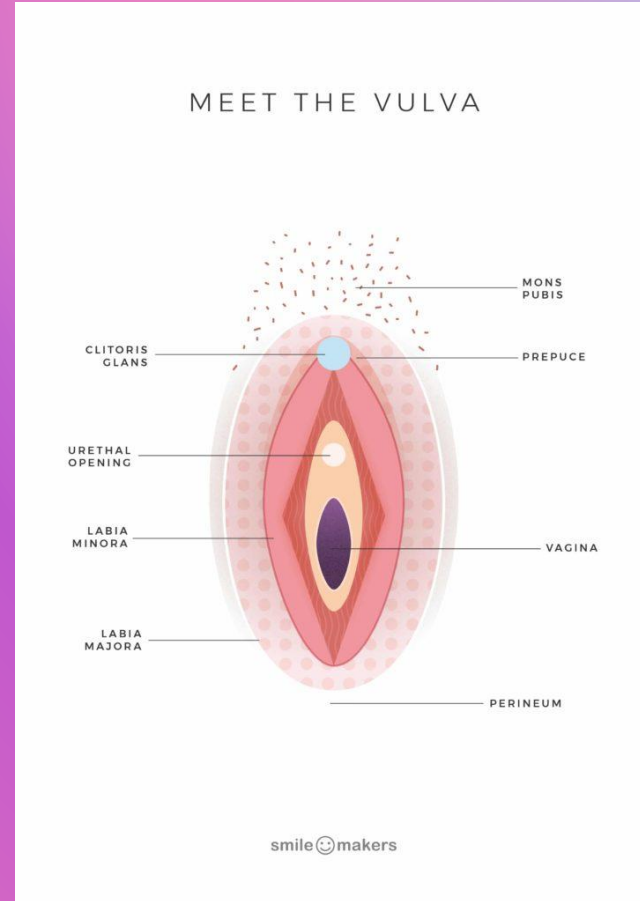


Pheromones and Their Role in Sexual Behavior

Pheromones — chemical signals that elicit behavioral or psychological responses in others



External Female Structures (Vulva)





External Female Sex Organs

Vulva — the external female genitals; often incorrectly called the vagina.

Mons pubis (mons veneris) — the fatty tissue covering the pubic bone.

Pubic bone — part of the pelvis; covered by the mons pubis.

Labia majora— the outer lips of the vulva.

- Extends from the mons pubic and enclose the labia minora, clitoris, urethral opening and vaginal entrance

Labia minora — the inner lips of the vulva, one on each side of the vaginal opening.





External Female Sex Organs

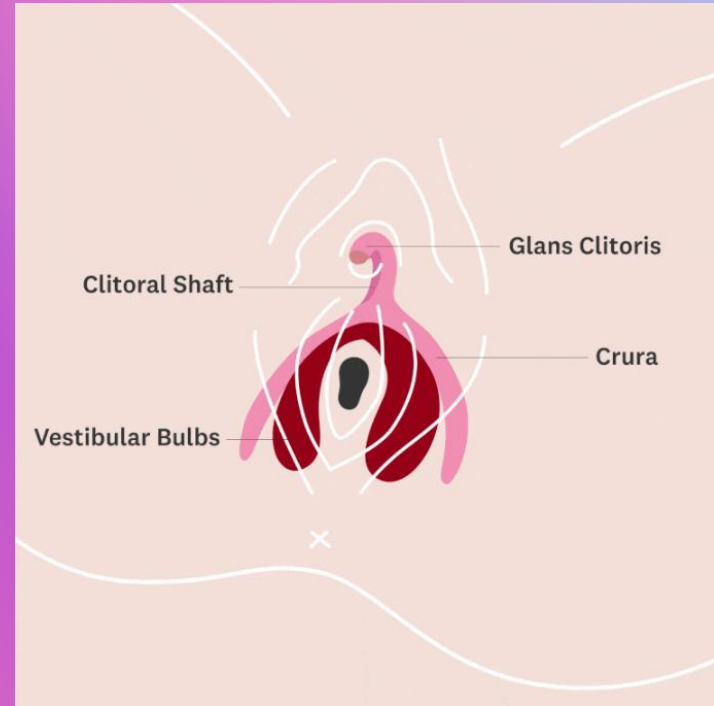
Sebaceous glands — glands that produce oil.

Sebum — oil from sebaceous glands that coats and protects the tissue surface

Tumescence — the state of being swollen or engorged with blood.



External Female Sex Organs





External Female Sex Organs

Clitoral hood — fold of skin that surrounding and protecting the clitoral glans.

Clitoris — highly sensitive structure of the female external genitals.

Clitoral glans — small external portion of the clitoris.

Clitoral crura — internal portion of the clitoris.





External Female Sex Organs

Urethral opening — opening between the clitoris and vaginal opening; for urine release

Vaginal opening — opening between urethral opening and perineum; leads to the uterus.

Anus — opening of the digestive tract; for feces release

Perineum — sexually sensitive area in both sexes

- Between vulva and anus
- 



External Female Sex Organs

Mucous membrane — inner wall of the vagina; made of moist, soft tissue.

Bartholin's glands — glands located on each side of the vaginal opening.

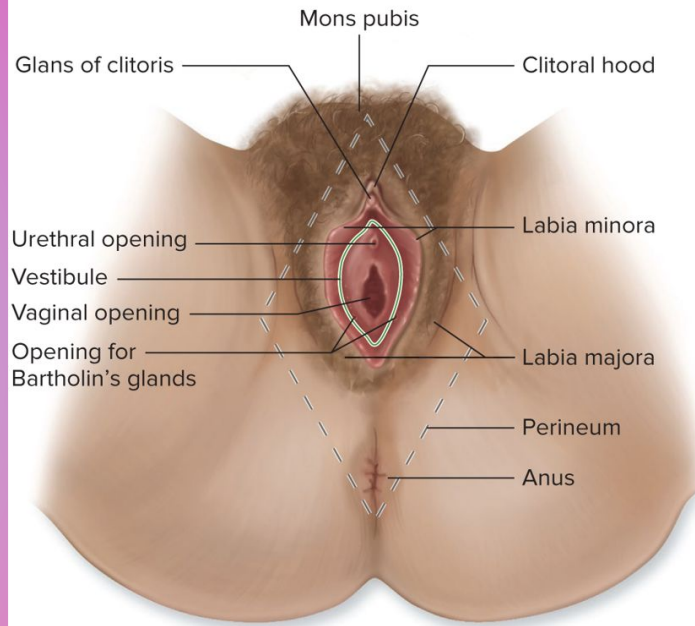
Tenting effect — expansion of the vaginal wall's during sexual arousal.

Vaginal lubrication — fluid from Bartholin's glands that aids penetration during sexual activity.

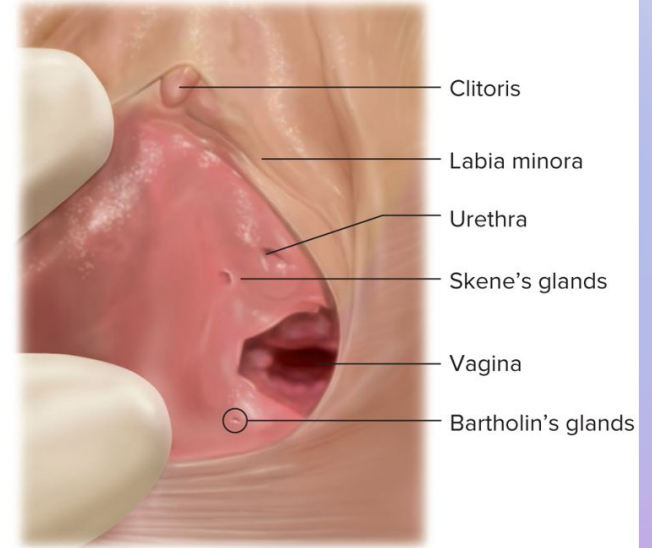


Figure 4.1

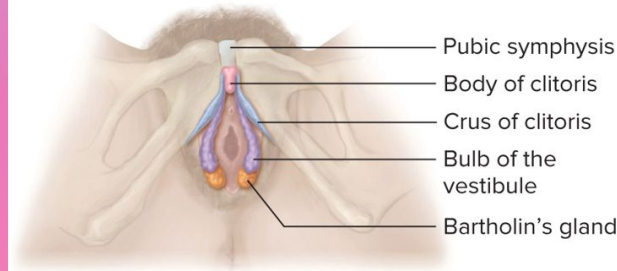
The structures of the vulva, the external female genitals



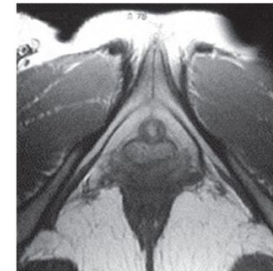
(a) Superficial structures of external genitalia



(b) Closeup of clitoris, Skene's glands, and Bartholin's glands



(c) Deep structures of external genitalia



(d) Scan of internal structures of clitoris



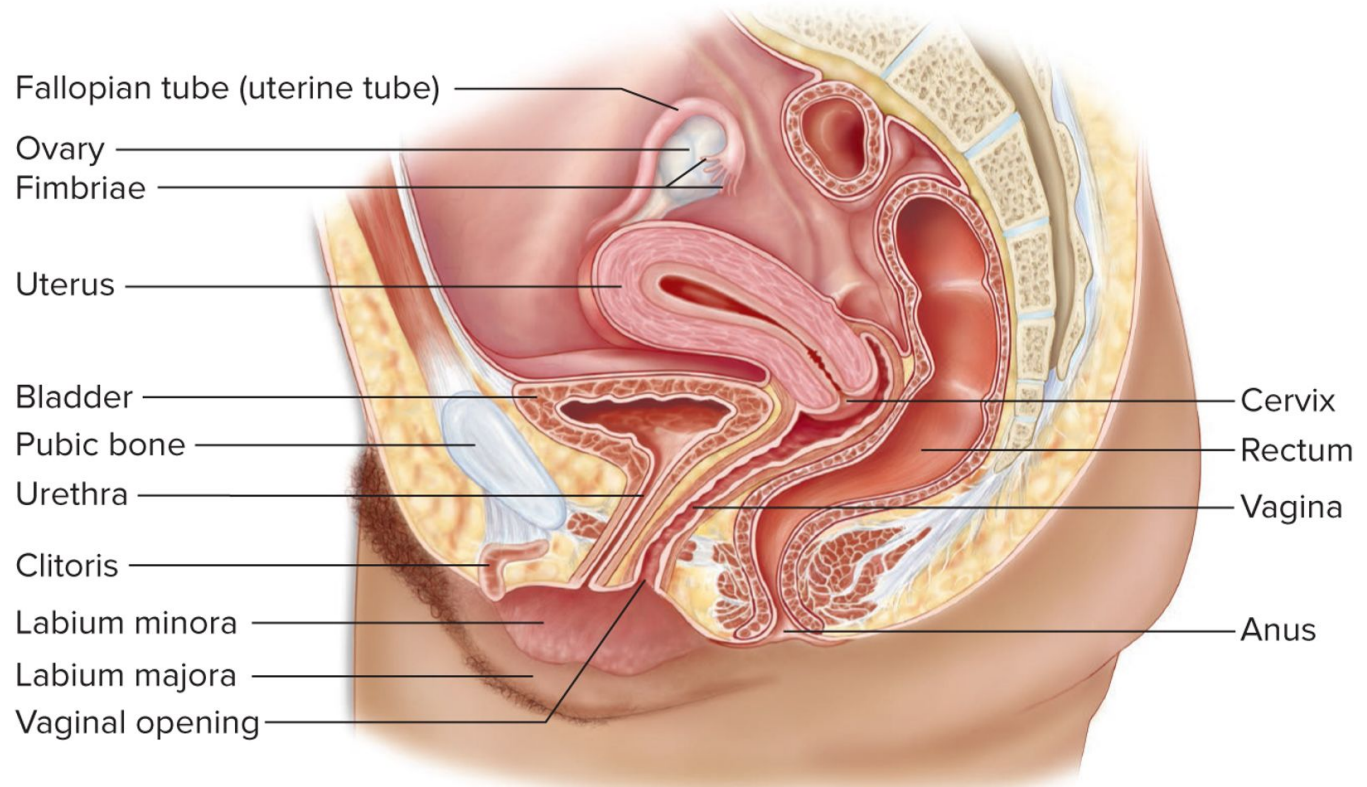
Internal Female Sexual Organs

Vagina — elastic, muscular canal extending from the vulva to the cervix

Cervix — neck of the uterus connecting it to the vagina



Figure 4.4
Side view of
the female
reproductive
system





Internal Female Sexual Organs

Uterus — connected to the cervix and both fallopian tubes; also called the womb

Three layers

- Perimetrium — outer uterine layer
 - Myometrium — middle muscular and vascular layer
 - Endometrium — inner lining, shed during menstruation
- 



Internal Female Sexual Organs

Fallopian tubes — ducts connecting the ovaries to the uterus

Fimbria — fringe tissue near the ovary that leads to the fallopian tube

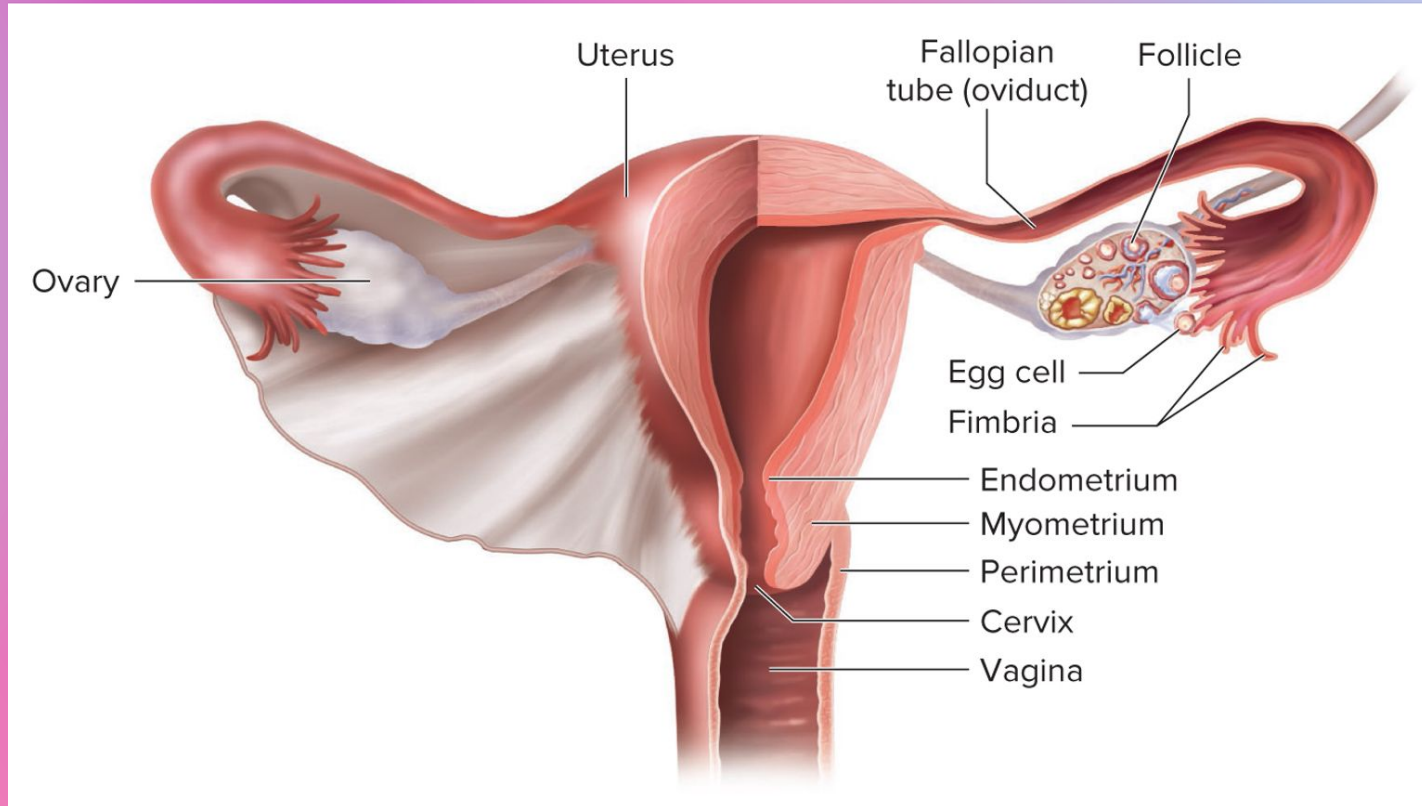
Ovary — contains gonads and endocrine glands; connects to fimbria to release an ovum into the uterus

- Most females have two ovaries

Ovarian follicles — hollow ball of cells containing an immature egg



Figure 4.6
Internal
front view of
the ovary,
fallopian
tube, and
layers of the
uterus





Internal Female Sexual Organs

Introitus — opening of the vagina

Hymen — fold of tissue partially covering the introitus

- Imperforate hymen — hymen completely closes the vaginal opening, causing menstrual fluid to accumulate
- 

Types of hymen

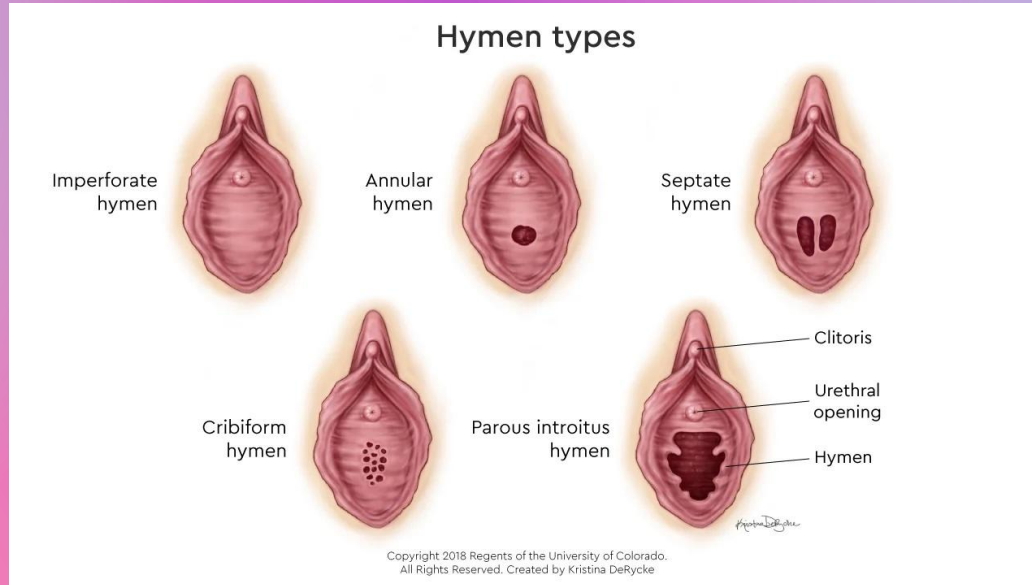
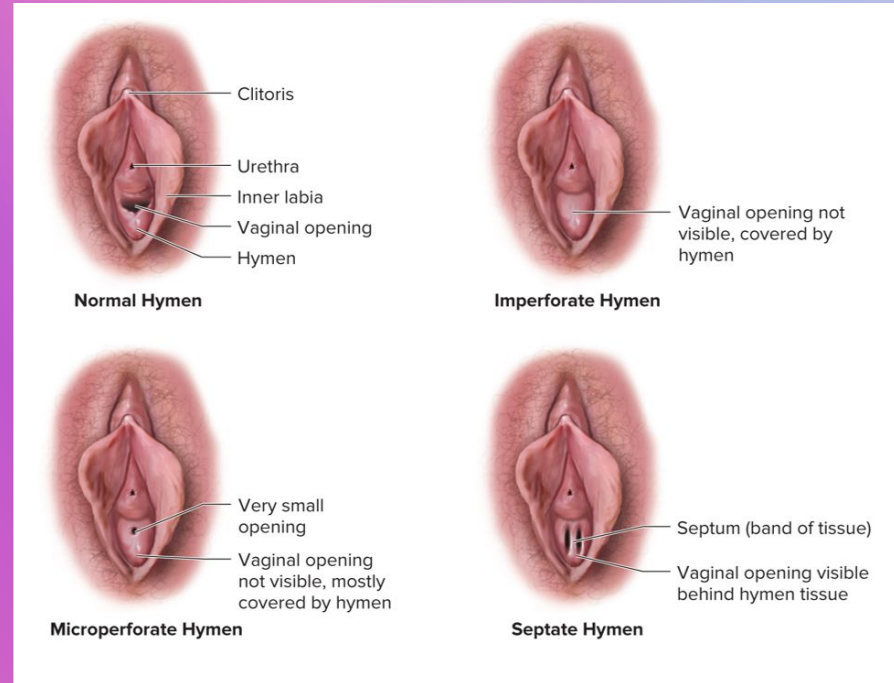


Figure 4.5 The hymen

The hymen, like other sexual structures, varies in shape and size among females. Imperforate hymens are rare and can be corrected with minor surgery. A microperforate hymen has a very small opening, and a septate hymen is divided in two by a band of tissue. Microperforate and septate hymens can be corrected with minor surgery.





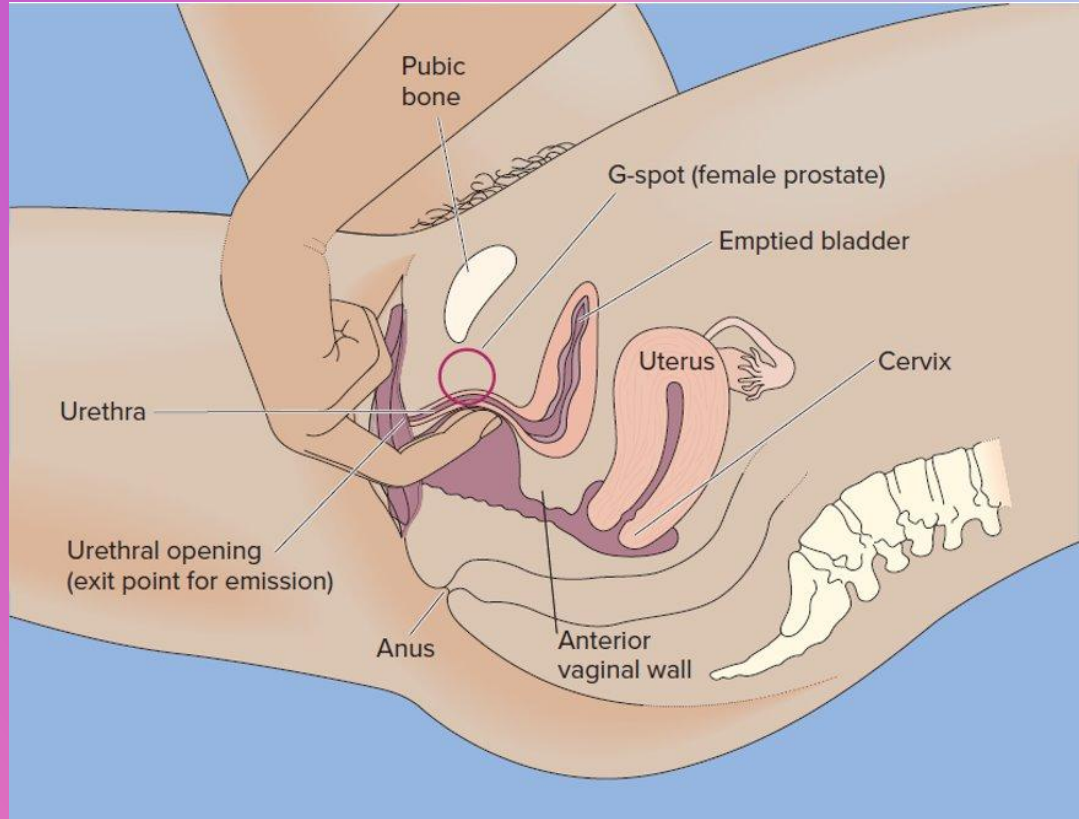
Internal Female Sexual Organs

G-spot (Grafenberg spot) — potentially erogenous area of the vagina thought to produce powerful orgasms

Vasocongestion — increased blood flow to erectile tissues in the genitals



The Grafenberg spot





Breasts and Breast Development

Secretory glands — form at the ends of the breast's milk ducts

- **Mammary glands** — internal breast tissue; produce milk to nourish an infant

Areola — dark, visible center of the breast

Alveoli — glands that start milk production after childbirth

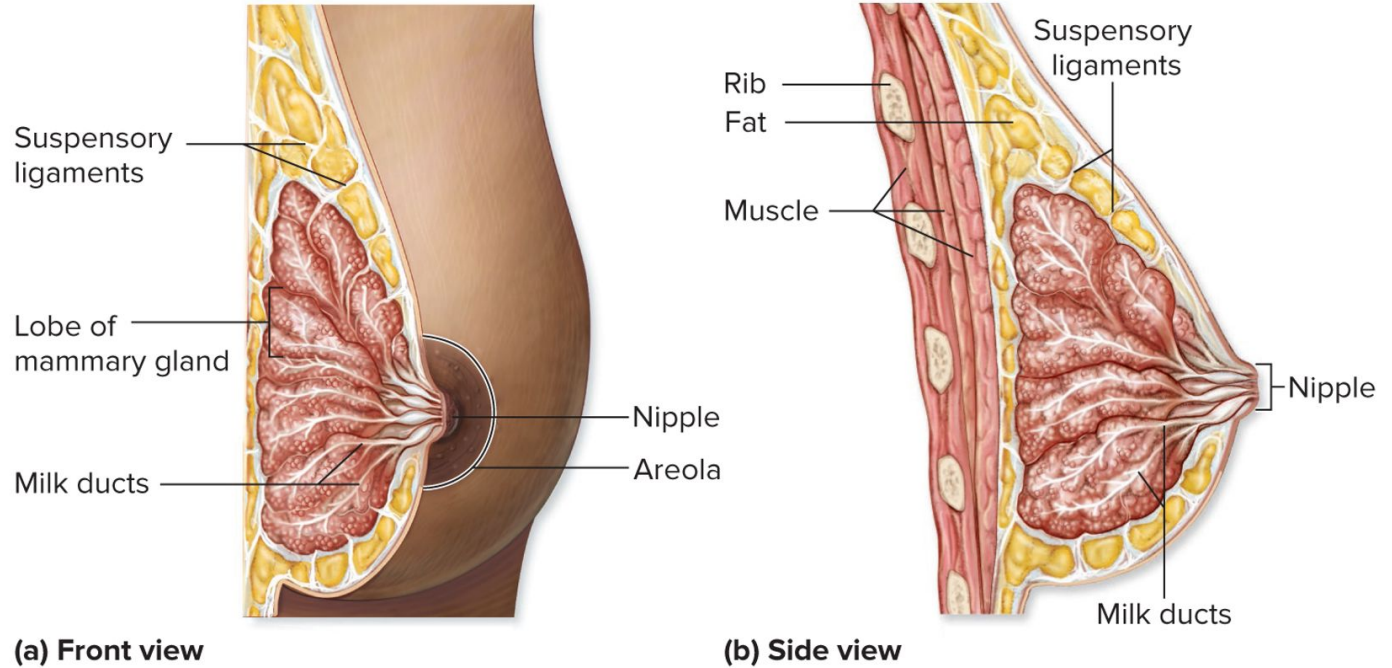
Nipple — raised bud at the center of the areola

Thelarche — first stage of breast development



Figure 4.7

Internal and external structures of the female breast





Variability in the Female Body

- All bodies vary in some way.
 - Differences in size, shape, and color
 - Bodies also vary anatomically.
 - Stress
 - Not signs of dysfunction
 - Beauty standards vary (Chapter 12)
- 



Female Genital Mutilation

Female circumcision — removal or shortening of the clitoris

- Labia suturing — prevents intercourse or hymen rupture

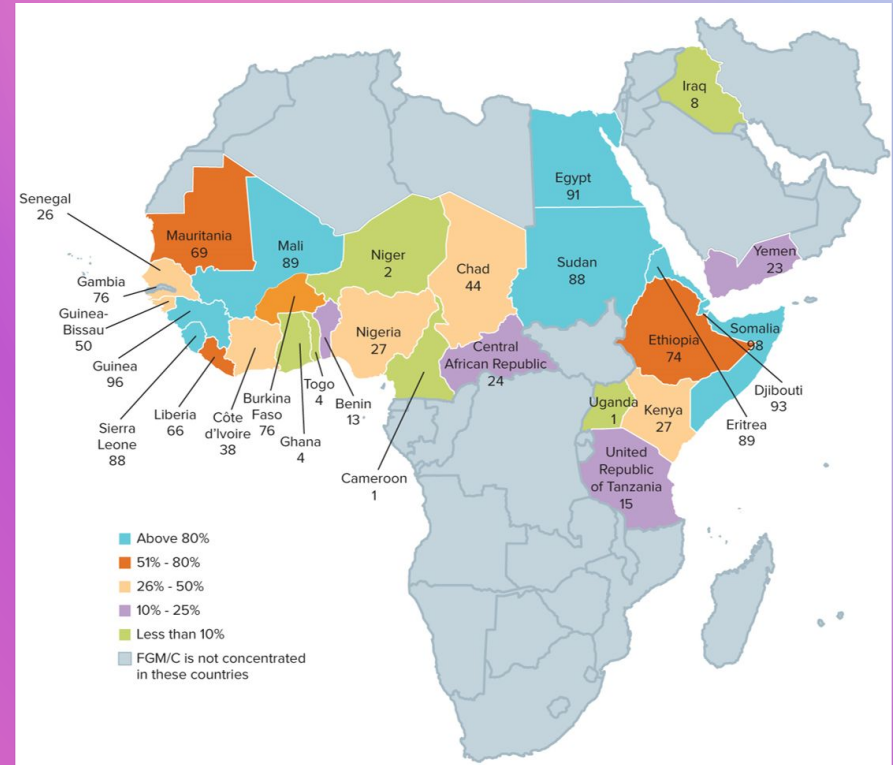
Female genital mutilation (FGM) — female circumcision without antiseptic or anesthesia

- Associated with the transition from girlhood to womanhood
 - Highly controversial cultural practice
- 

Figure 4.9

Distribution of female genital mutilation FGM in Africa

This map illustrates the distribution of FGM: the percentage of the female population that is subject to FGM and by country.





The Uterine vs Ovarian Cycle

Ovarian — refers to the ovaries and the cycle in which the egg develops

Uterine — refers to the uterus and the cycle in which the womb is prepared for pregnancy





The Ovarian Cycle

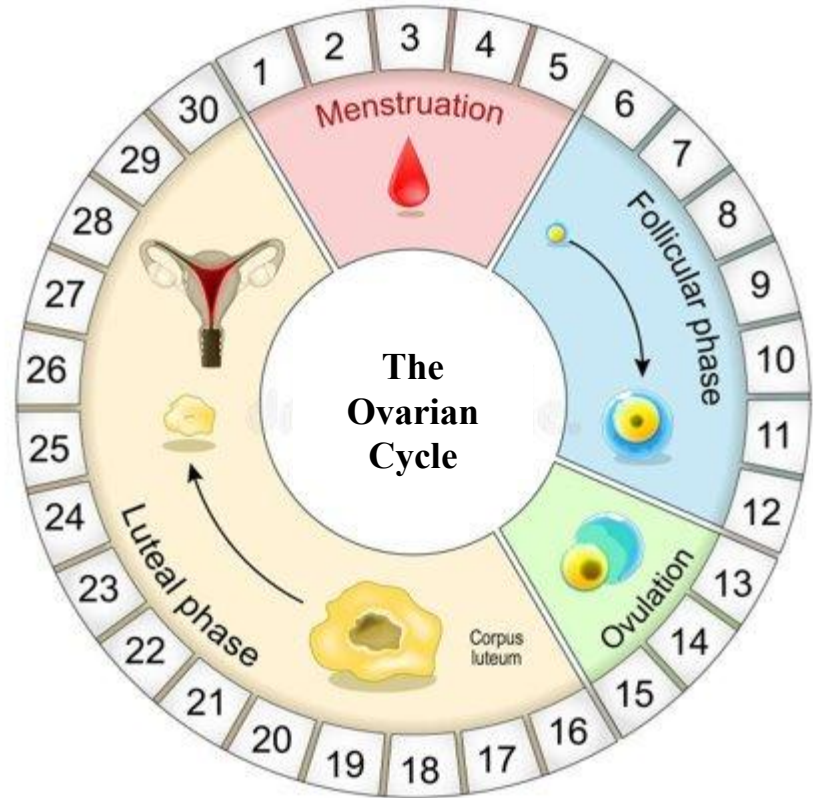
Oogenesis — “egg beginning”

- About 400 oocytes mature and are released in a woman's lifetime, starting at puberty
- Women are born with a fixed number of oocytes, unlike men who produce sperm throughout life

Four phases

- **Follicular** — ovarian follicles develop; one ovum matures
 - **Ovulatory** — mature ovum is released from the ovary (ovulation)
 - **Luteal** — corpus luteum forms; secretes hormones to support potential pregnancy
 - **Menstrual** — hormone levels drop if no fertilization occurs; ovum and uterine lining are shed
- 

The Ovarian Cycle





The Uterine Cycle

The uterus prepares for implantation

Three phases:

- **Proliferative** — endometrium thickens; ovum matures
 - **Secretory** — endometrium thickens more; corpus luteum develops
 - **Corpus luteum** — tissue from ruptured follicle
 - Corresponds with the luteal phase of the ovarian cycle
 - **Menstrual** — ovum passes through the fallopian tube into the uterus
- 



The Uterine Cycle

Menstruation — shedding of the uterine lining (“woman’s period”)

Menses — menstrual flow, typically 3–7 days

Menarche — first menstrual period

- Average age in the U.S. is 12 -14; has declined historically

Menstrual synchrony — alignment of menstrual cycles among women who live together



The Uterine Cycle

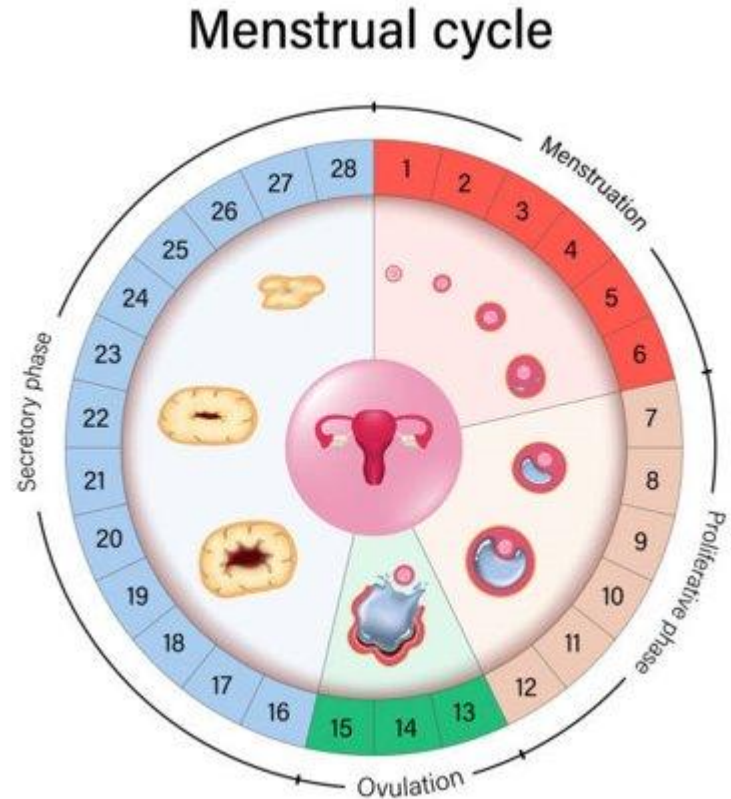
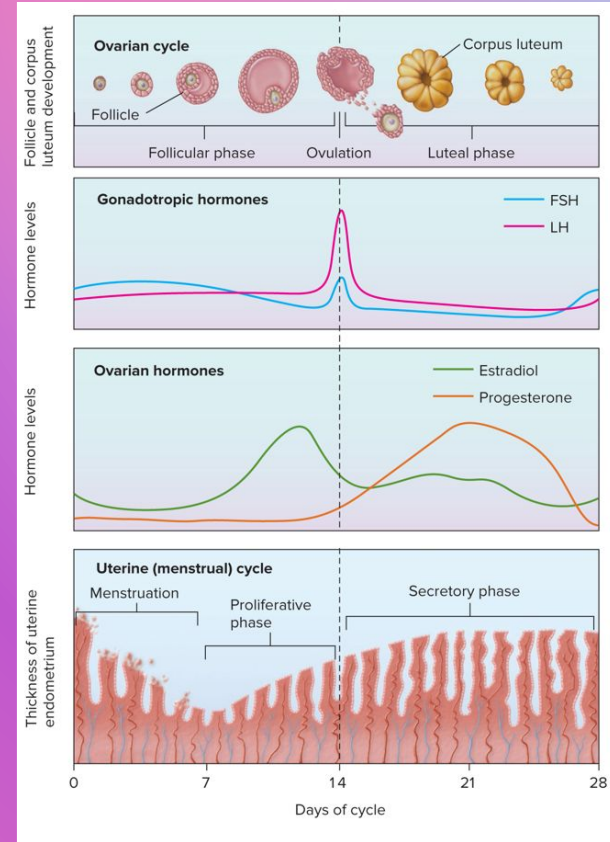


Figure 4.10 The three phases of the menstrual cycle

F S H stimulates the maturation of an ovum, and L H triggers the release of the ovum. If the ovum isn't fertilized and implanted in the uterine wall, it dissolves, and the endometrium is shed during menstruation.



Menstrual products





The Menstrual Cycle

Most women experience physical, emotional, or behavioral changes in the week before menstruation

Premenstrual syndrome (PMS) — collection of physical and emotional symptoms before/during menstruation

- Neurotransmitter fluctuations may play a role

Premenstrual dysphoric disorder (PMDD) — severe depression, irritability, and tension before menstruation





The Menstrual Cycle

Menorrhea — very heavy or prolonged menstrual bleeding

Dysmenorrhea — severe uterine pain during menstruation

Amenorrhea — absence of menstrual periods





The Menstrual Cycle

Toxic shock syndrome (TSS) — severe disease causing fever, shock, and organ dysfunction

- Caused by bacterial toxins and can be fatal if untreated
 - Risk factors — childbirth, staph infection, menstruation, surgery, tampons, barrier contraceptives
 - Symptoms — confusion, diarrhea, headache, fever, low BP, muscle aches, nausea, organ failure, rash
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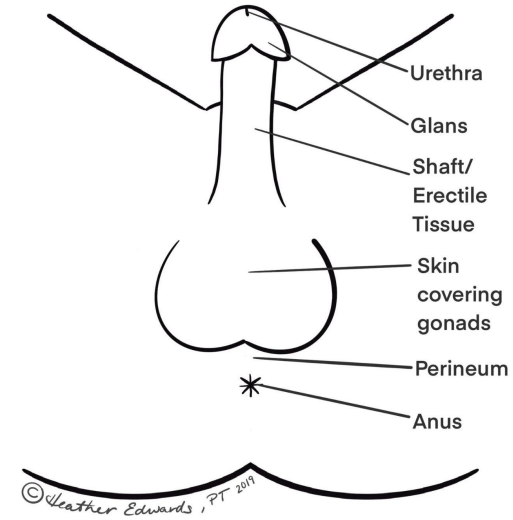


The Menstrual Cycle

Contact a health care provider if:

- Menarche hasn't occurred by age 16
 - Menstrual periods suddenly stop
 - Menstrual bleeding is excessive
 - Severe pain occurs during menstruation
- 

External Male Sex Organs





External Male Sex Organs

Penis — primary male organ for sexual pleasure; transports semen and urine

- **Root** — base
 - **Shaft** — part of the penis between the glans and the body
 - **Head** — tip
-
-



External Male Sex Organs

Penile glans — sexually sensitive area at the top of the penis with many nerve endings

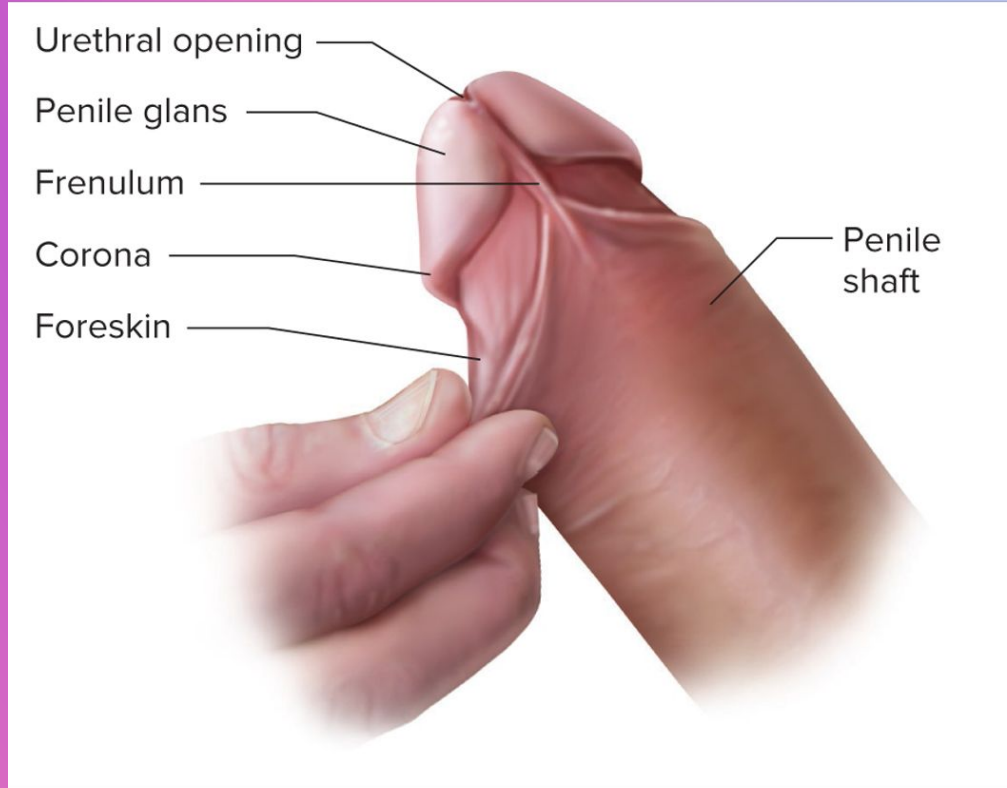
- **Corona** — rounded area at the base of the glans

Foreskin — thin, sensitive skin covering part of the shaft, corona, and glans

- **Smegma** — cheesy substance from secretions that collects under the foreskin

Frenulum — sensitive tissue on the underside connecting glans, shaft, and foreskin

Figure 4.12 The external structure of the penis





External Male Sex Organs

Scrotum — extension of the abdomen between the penis and anus

Scrotal sac — holds the testicles

Testicles (testes) — male gonads; produce sperm and testosterone

Perineum — sexually sensitive area in both sexes

- Between scrotum and anus

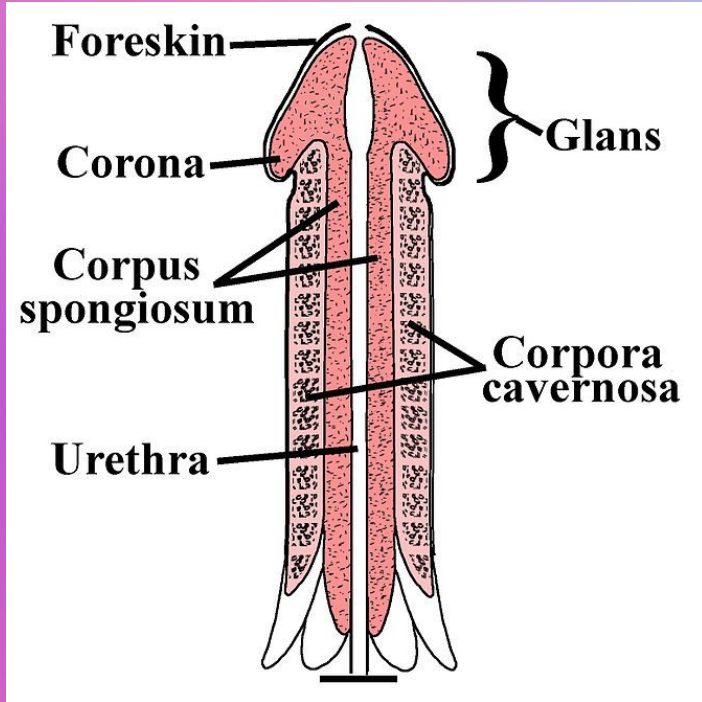


Internal Male Sex Organs

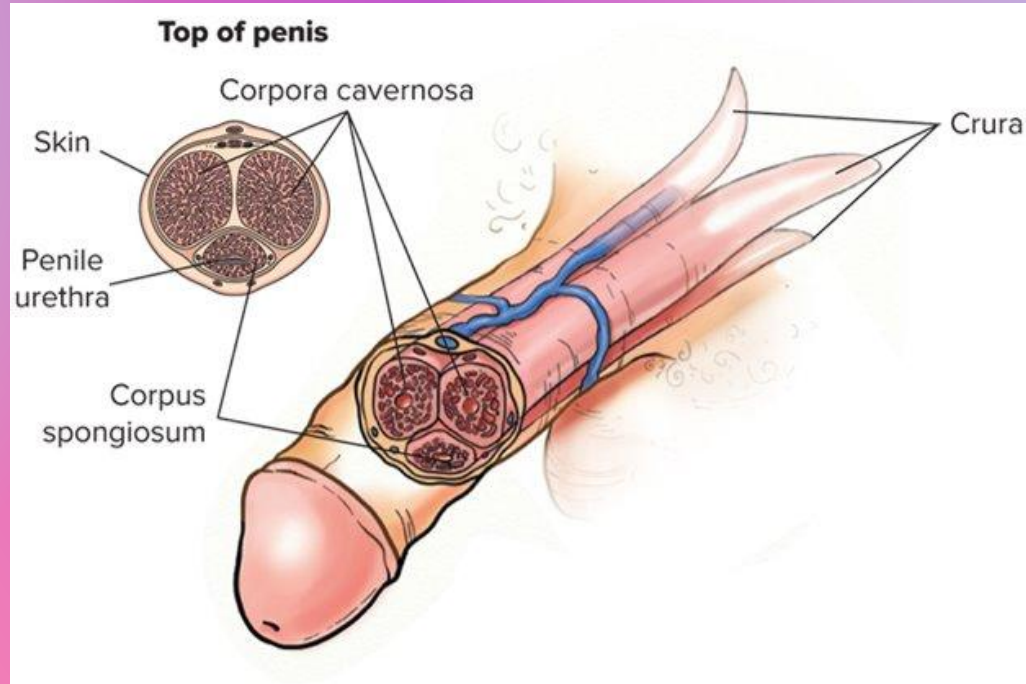
Internal structures of the shaft

- **Corpus cavernosa** — two paired cylinders that fill with blood during arousal, causing an erection
- **Corpus spongiosum** — spongy tissue surrounding the urethra inside the penis

Internal Male Sex Organs



Interior Cross section of the Penis





Internal Male Sex Organs

Cremaster muscle — in the scrotum; helps maintain ideal temperature for sperm production

- Causes the “spasm”

Seminiferous tubule — thin, coiled structure in the testes where sperm is produced

Tunica albuginea — capsule that holds the seminiferous tubules



Internal Male Sex Organs

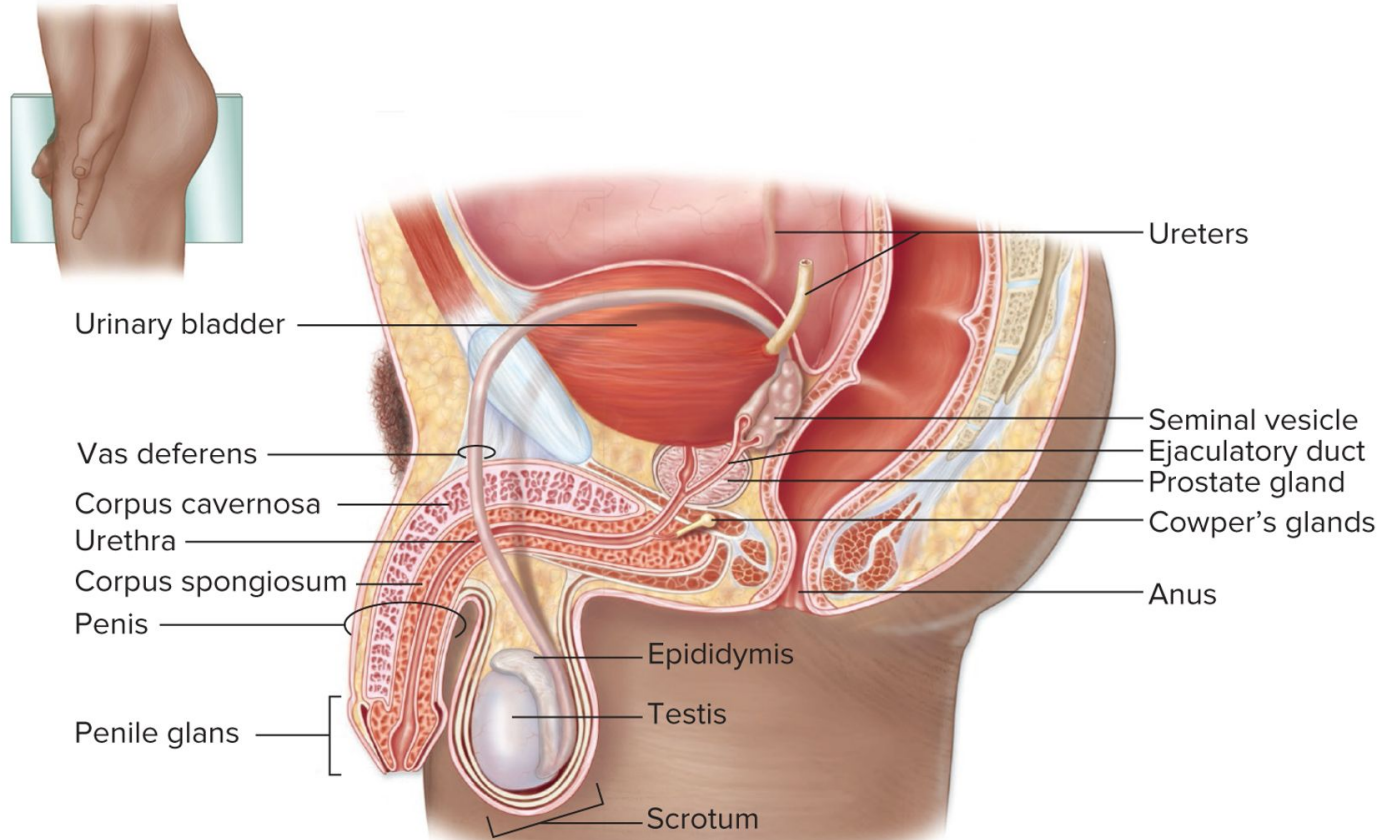
Epididymis — tightly coiled tube atop each testicle

Vas deferens — ducts that transport sperm from the epididymis to the ejaculatory ducts

Figure

4.11

Side view of
the internal
structures
of the male
genitals



**Figure
4.13**
Internal
view of the
testicles
inside the
scrotum

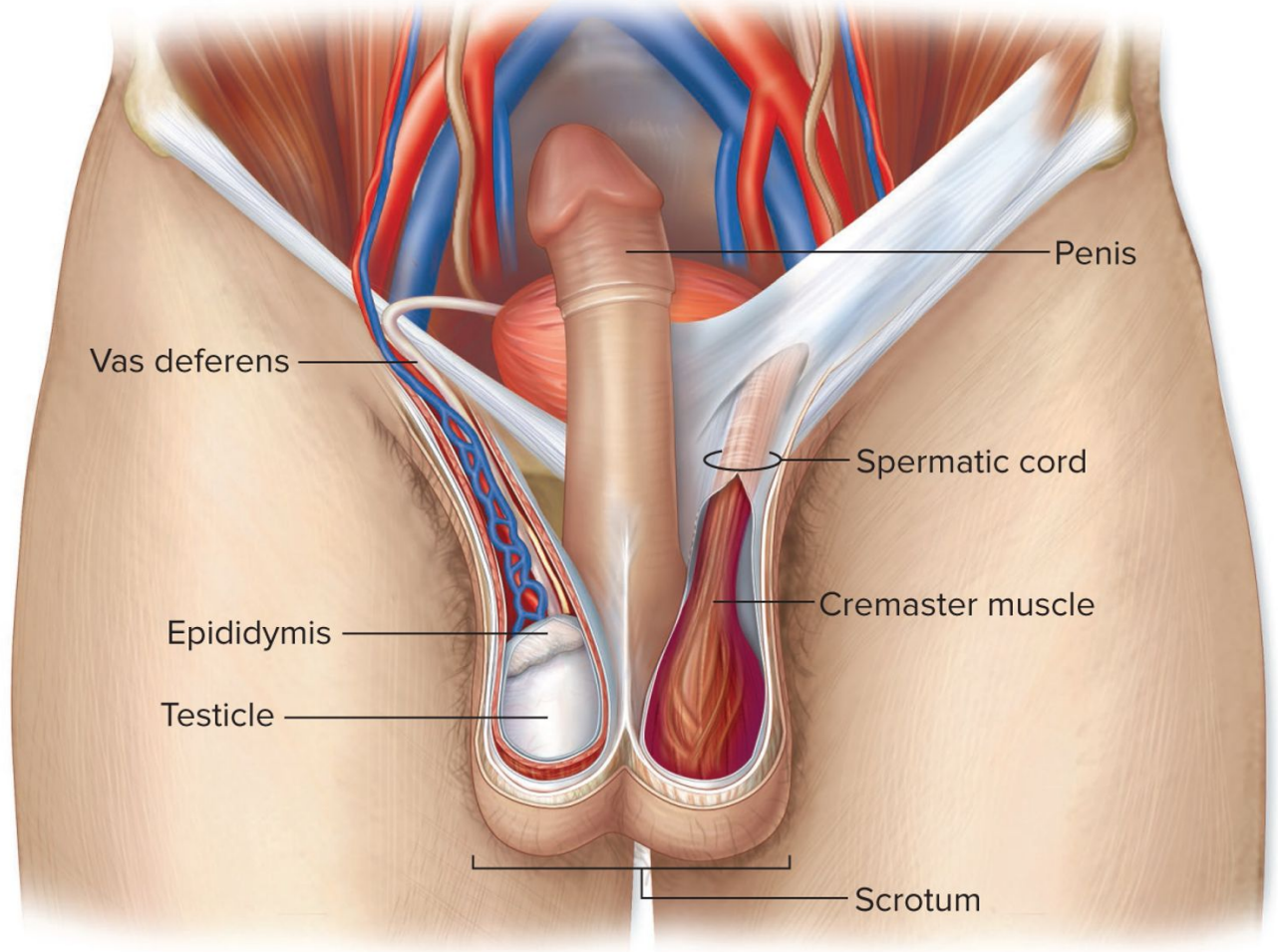
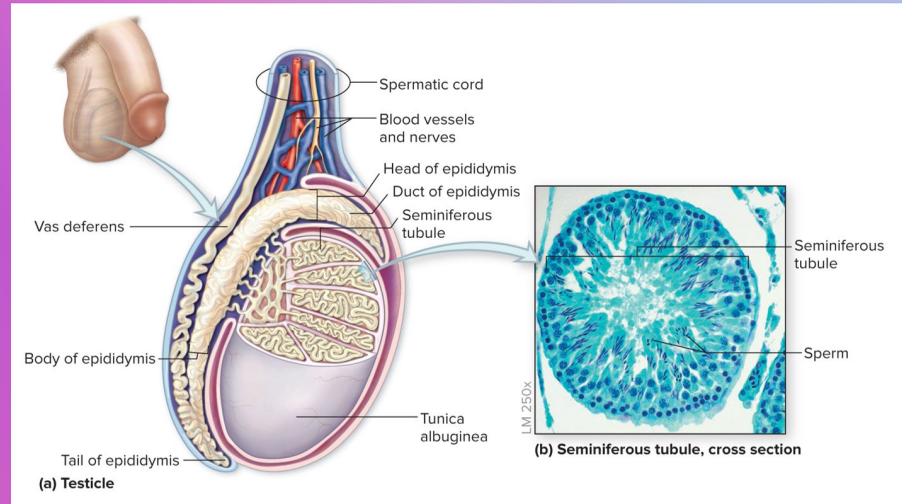


Figure 4.14 Testicles and seminiferous tubules

(a) Cut-away view of a testicle. (b) Greatly enlarged photomicrograph of a seminiferous tubule in cross section. An unfolded seminiferous tubule would stretch over one mile.





Variability in the Male Body

Male body varies in shape, size, color, function

- Unaroused (Flaccid) penis — average is slightly under 3 inches, although great variation
 - Greatest difference is in flaccid state
- Erect penis — mean length 5.5 inches, determined by hereditary factors
 - Erection causes them to be uniform

Size ≠ main factor for sexual pleasure

Figure 4.15 Penis shapes

The size and shape of the penis vary from one man to another. Note that the penis on the left is uncircumcised.



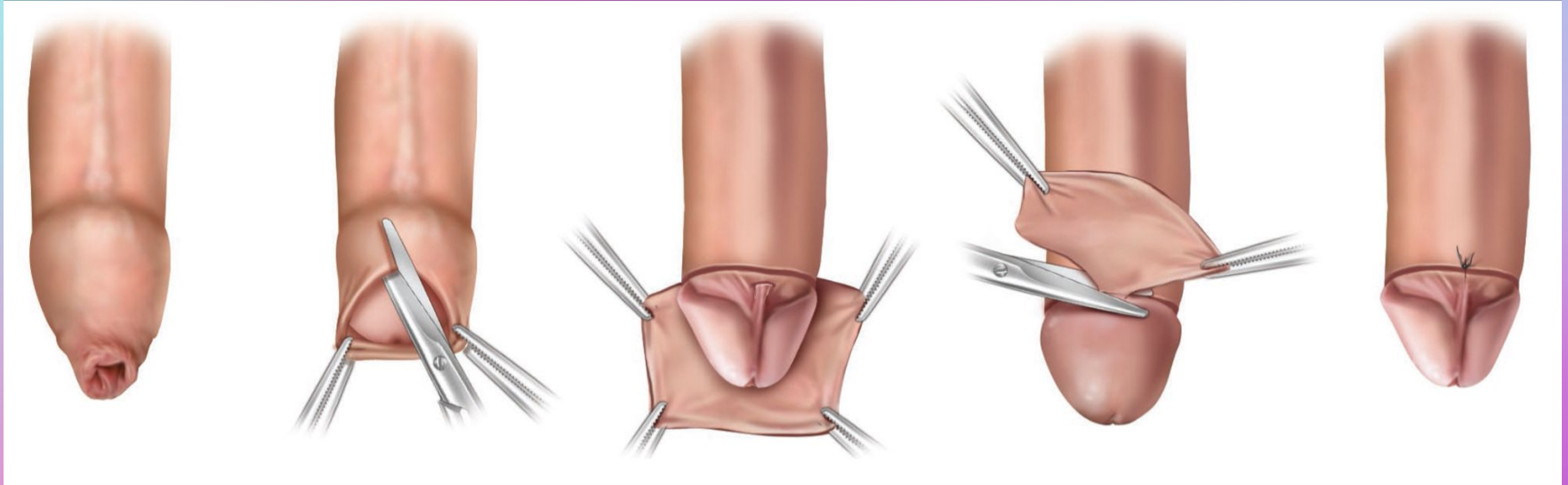


Circumcision

Male circumcision — removal of the foreskin; commonly performed on infant boys

- Benefits may reduce urinary tract infections, HIV and some STI risk, and penile cancer
 - Human rights advocacy groups recommend males to make the decision themselves
-

Figure 4.16 The steps involved in the male circumcision procedure





Male Sperm Production

Spermatogenesis — process of sperm production





Erection and Ejaculation

Penile columns (erectile tissues) — spongy tissue in the penis that fills with blood during an erection

Erection — firm, enlarged state when erectile tissue fills with blood

Ejaculation — ejection of semen (usually containing sperm) from the male reproductive tract, usually with orgasm

- Controlled by the autonomic nervous system
- Epididymis → vas deferens → urethra



Erection and Ejaculation

Semen — sperm + fluids from prostate, seminal vesicles, Cowper's glands

- **Seminal vesicles** — produce majority of semen
- **Prostate gland** — produces one-third of seminal fluid
- **Cowper's gland (bulbourethral gland)** — secretes clear, mucus-like fluid during arousal
 - Pre-ejaculate (pre-cum) mucus



Erection and Ejaculation

Ejaculation occurs in two stages:

Emission phase — cremaster muscle pulls testes close to the body, preparing sperm release

- **Ejaculatory inevitability** — point during intercourse when semen is about to be released; male feels “about to come”

Expulsion phase — rhythmic contractions force semen out of the body

- **Ejaculatory ducts** — two short ducts in the prostate that transport sperm to the urethra before ejaculation



Erection and Ejaculation

Spermarche — first ejaculation; often occurs during sleep (nocturnal emission)

Detumescence — softening of the penis after ejaculation

Retrograde ejaculation — semen ejaculated into the bladder

Nocturnal emission — appearance of semen during sleep



Bodily Integrity Rights and Sexual Well-Being

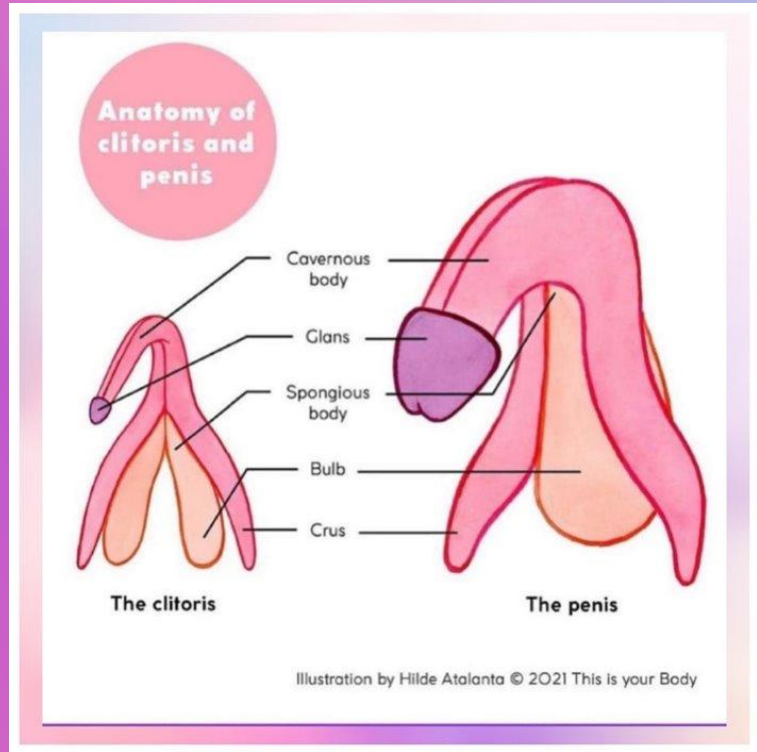
Bodily integrity — the right to personal autonomy and self-determination over one's body

Genital integrity — the idea that genitals should remain intact and uninfluenced by others

Cultural differences make these issues controversial

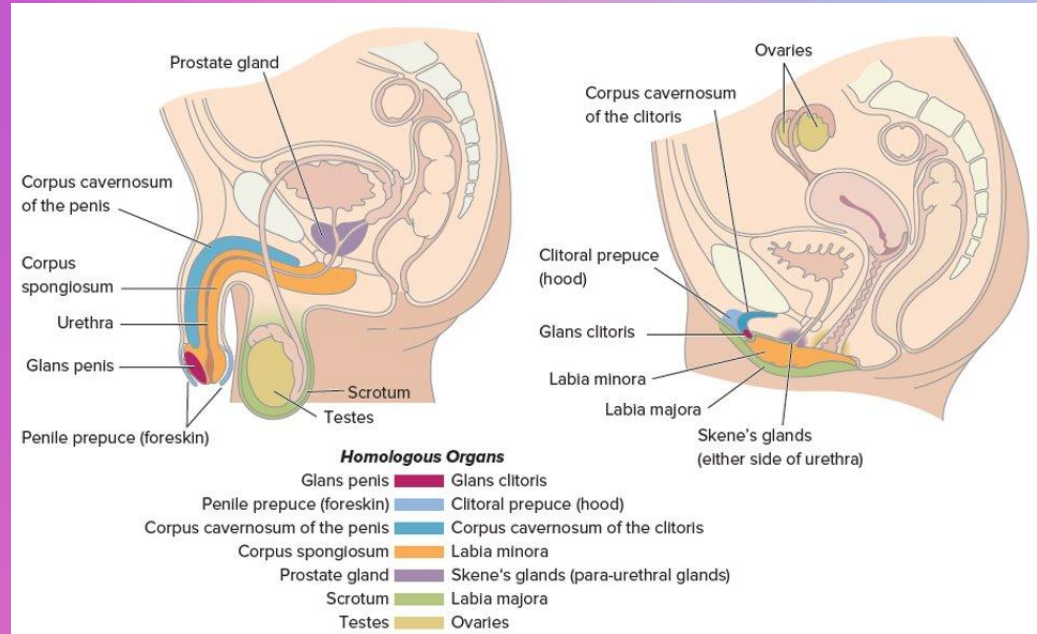
Fetal Differentiation of

We Really Aren't That Different



Homologous Organs

Each male sexual structure has a homologous, or similar characteristic, developed from the same cells in developing female fetus

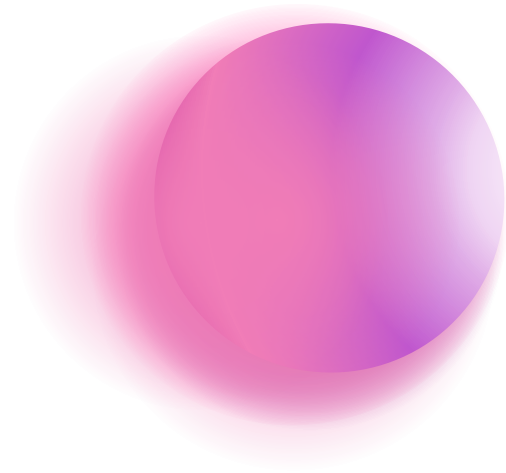


Activity: Unlabeled figures

Groups: 4–5 students

Instructions: Use the word bank to identify and label the following structures. Access the document titled "Sexual Anatomy and Physiology Unlabeled Figures" in Canvas to complete this activity together.

DO NOT USE YOUR NOTES



Questions?